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Research statement
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My primary research interests lie in model design and numerical investigations using Quantum Monte Carlo (QMC) methods, with a focus on quantum criticality, non-Fermi-liquid behavior, and strongly correlated moiré systems. I have also recently been working on Kondo problems, Kagome metal and Entanglement Entropy in interacting fermion systems.

In addition, I am interested in algorithmic developments and in understanding the fermion sign problem in QMC simulations. More broadly, my research is organized around four major themes that guide and unify my work:

1. Kondo physics and gauge-field–coupled systems

I am developing model for Kondo systems coupled to $U(1)$ gauge fields, an important setting for heavy-fermion[1]. Traditional Kondo-Heisenberg models involve highly complex composite operators, leading to prohibitive Wick expansions and unreliable simulations. To overcome this, we constructed a hybrid model of conduction electrons c , localized fermions f , and compact $U(1)$ gauge fields, enabling sign-problem-free QMC.

This formulation incorporates a π -flux Dirac dispersion and a composite fermion $d = zf$ that naturally encodes Kondo hybridization. Using this method, I computed composite-fermion spectral functions and optical conductivity across the Kondo to Kondo-breakdown transition. Hybridized heavy bands appear in the Kondo phase and vanish in the breakdown regime, providing a clear numerical signature of the transition. In addition, the change in optical conductivity further reveals clear evidence for its interpretation as an orbital-selective Mott transition. This work opens a new pathway for studying Kondo breakdown and gauge-field using unbiased QMC.

2. Correlated and topological phases in Moiré and Kagome systems

Moiré superlattices and Kagome lattices—have emerged as ideal platforms for studying flat-band physics, strong correlations, and unconventional ordering. My contributions span both methodological advancements and new physical insights.

Quasiparticle Spectroscopy of Chiral Charge Order:

My recent collaborative work on kagome metals is built upon the mean-field model and spectroscopic framework established in our study of chiral charge order in CsV_3Sb_5 .

In this project[2], we combined numerical modeling with STM quasiparticle spectroscopy to reveal how a local magnetic impurity couples to the loop-current sector of the chiral flux phase. Our calculations demonstrated that the magnetic moment of a single Co atom locally

reverses the circulating current on a kagome plaquette, thereby generating a topological in-gap quasiparticle excitation inside the CDW pseudogap—an effect fully consistent with STM observations. This work provides microscopic evidence supporting a time-reversal-breaking loop-current ground state, distinguishes it from competing charge-bond-order(CBO) scenarios, and establishes a general paradigm for probing flux-sector excitations in correlated Kagome systems.

Momentum-space models and quantum Monte Carlo method:

We start from the continuous Bistritzer-MacDonald(BM) model and long range single-gate Coulomb interaction, and integrate out the states on the remote bands to obtain the projected Hamiltonian, which has considered the long-range Coulomb interaction and the Wannier obstruction problem. The momentum-space method [3] we developed is free of the sign problem at charge-neutrality point(CNP) with the C_2P and C_2T symmetries. For other integer filling, there are sign problems in general, but we can also discussed the computational possibilities[4].

Twisted Bilayer Graphene and Twisted TMD:

Since we developed the momentum-space QMC method for twisted system. We compute the spectra of both single-particle and particle-hole excitations. We found that at CNP [5], the intervalley coherent(IVC) state is the leading instability, with strong competition from the valley polarized (VP) state. In addition, at low-energy, longlived valley waves are observed in close analogy to spin waves of a Heisenberg ferromagnet, while these modes become over-damped as their energy reaches the particle-hole continuum.

And in the chiral limit and flat band limit, it is fortunate enough for us to be able to study the physics of $\nu = 1$ [6], although there is a sign problem now. In the low temperature limit, the sign of this system converges to a finite value and decays algebraically rather than exponentially with system size L . This allows us to perform QMC simulations. The Ising thermal transition and the QAH-TMI state which emerges at $T = 0$ are observed. The transition temperature T_c is dramatically smaller than the energy scale of the gap at zero temperature. And we show that it's caused by excitonic excitations in the particle-hole channel.

We also reveal nontrivial phenomena in twisted TMD[7], such as doping independent gap and large compressibility above the superconducting dome.

3. Quantum Monte Carlo method development: algorithms, entanglement, and the sign problem

As mentioned earlier, we designed a lot of models[8, 9, 10, 11, 3, 5, 12, 13, 14, 15, 16] without sign problem to conduct different studies. At the same time, we have also carried out some tests[17] for different QMC update algorithms and also study and develop the algorithm of Rényi entanglement entropy.

In addition, in order to deepen our understanding of the sign problem in QMC[18, 19], especially the fermion sign problem, we develop the Sign Bound Theory.

Algorithms for Entanglement entropy:

We examined the theoretical and numerical structure of methods used to compute Rényi entanglement entropy in interacting fermion systems[20], and on this basis developed a significantly more stable integral algorithm for evaluating exponential observables, including entanglement entropy and free energy[21, 22]. Furthermore, by introducing an annealing-inspired improvement, we enhanced the efficiency of the method for obtaining entanglement entropies as continuous functions of parameters[23].

Sign Problem in QMC:

People often expect a connection between the Monte Carlo sign and phase transitions. In our work[24], we investigated and conjectured possible relations, and clarified that extrema of the sign or its derivatives do not necessarily coincide with the actual transition points. We further proposed a modified sign quantity that corresponds rigorously to the phase transition.

Sign Bound Theory:

In addition, usually, if there is a sign problem, the average of sign usually decays exponentially to zero with inverse of temperature β and system size L . However, in some systems, we find that in the zero temperature limit, the sign does not decay exponentially with respect to system size L but rather decay algebraically.

In fact, in the zero temperature limit, the mean of the signs is related to the ground state energy and ground state degeneracy of the original system and the reference system. When we choose a good reference system, the two ground state energies are equal, and the sign is just a function of the degeneracy of the two systems. Our Sign Bound Theory[19, 4] discusses the relevant details and attempts to give an upper bound on the degeneracy of the reference system in some cases, which leads to a lower bound on the sign.

The theory guides us to QMC simulations of some twisted systems[4, 6] with sign problem. In these cases, although there is a sign problem, the average of sign has a polynomial lower bound with respect to system size L in the zero temperature limit, so the simulation can be carried out at an acceptable cost.

4. Other interests: Quantum criticality and nFL behaviors

Understanding non-Fermi-liquid behavior in itinerant electron systems is a central challenge in correlated electron physics. My work aims to clarify the universal features of nFL regimes and the microscopic mechanisms underlying critical behaviors.

Spin-fermion coupled model:

In a general context, nFL behavior often occurs via electron interactions mediated by gapless bosonic modes that render the electrons incoherent. Such gapless bosons typically arise in the vicinity of a QCP or in quantum gauge theories.

We designed a spin-fermion coupled square lattice model[8, 9], which has an itinerant QCP with fermion pockets and hot spots, and developed quantum Monte Carlo simulation to examine itinerant QCPs generated by antiferromagnetic fluctuations. Then we unam-

biguously reveal the nFL fermion self-energy and the bosonic dynamic susceptibilities.

Near the AFM-QCP, fermions at the hot spots shows non-Fermi liquid behavior. The Matsubara-frequency dependence of the $\text{Im}(\Sigma(k, \omega))$ goes to a small constant at low ω , and no sign of either linear vanishing or diverging are observed. The not a small anomalous dimension $\eta = 0.25$ we detected is significantly different from the Hertz-Millis theory prediction.

Self-tune spin-1/2 Yukawa-SYK model:

In addition, since it is very difficult to tune the system to the quantum critical region. The precise determination of the quantum critical point and region of nFL are subject to finite size effects, and the position of the QCP is not universal but system dependent. We also turn to study the self-tuned Yukawa-SYK model[10, 11].

Unlike the itinerant QCP models where the systems live in 2 spatial dimensions, the Yukawa-SYK models live in infinite dimensions and have been shown to “self-tune” to quantum criticality within large-N approximation. Here self-tuned means the system is critical, independent of the bosonic bare mass m_0 , which gives us a large area of nFL.

We performed unbiased sign-problem free quantum Monte Carlo simulations of the Yukawa-SYK model, and reported direct evidence of self-tuned quantum-critical and nFL behaviors of Green functions, which have the form $G_f(\tau, 0) \propto \left(\frac{\pi}{\beta \sin(\pi\tau/\beta)}\right)^{1-x}$ and $G_b(\tau, 0) \propto \left(\frac{\pi}{\beta \sin(\pi\tau/\beta)}\right)^{2x}$.

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